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Giuseppe Silano

Curriculum Vitae

Education

Nov 2020 **Ph.D. Program in Information Technologies for Engineering**, *Group for Research on Automatic Control Engineering (GRACE)*, University of Sannio, Benevento, Italy
- Additional mention of Doctor Europaeus.
- Focus on robotics, control, path planning, and software-in-the-loop.
- Supervisor: Prof. Dr. Luigi Iannelli.

Mar 2019 – Nov 2019 **Visiting Ph.D. student at Laboratoire d'Analyse et d'Architecture des Systèmes (LAAS), Robotics and Interactions (RIS) group**, *Centre National de la Recherche Scientifique (CNRS)*, Toulouse, France
- Control of fully-actuated 6DoFs[1] robots with onboard sensors.
- Supervisor: Prof. Dr. Antonio Franchi.

Mar 2016 **Master of Science in Electronic Engineering**, *University of Sannio*, Benevento, Italy
- Focus on robotics, control, electronics, and telecommunication.

Jul 2012 **Bachelor of Science in Computer Engineering**, *University of Sannio*, Benevento, Italy
- Focus on robotics, control, software, telecommunication, and electronics.

Feb 2012 – Jun 2012 **Industrial Internship in Systems Engineering**, *Mosaico Monitoraggio Integrato S.r.l.*, Benevento, Italy
- Design a software production methodology using an Object Oriented (OO) approach for Programmable Logic Controllers (PLCs).

Professional Affiliation

Dec 2016 – Today **IEEE (Institute of Electrical and Electronic Engineers)**, *Student Member (ST'17), Member (M'21), Senior Member (SM'25)*
- IEEE Control Systems Society (CSS).
- IEEE Robotics and Automation Society (RAS).
- IEEE Systems, Man, and Cybernetics Society (SMC).

[1]DoFs: Degree of Freedoms.

Academic Appointments

Jul 2022 – Today **Tenure Researcher**, *Department of Power Generation Technologies and Materials (TGM)*, Ricerca sul Sistema Energetico (RSE) S.p.A., Milan, Italy [RP1], [RP2], [RP3], [RP4]

Jul 2022 – Today **Associate Researcher**, *Multi-Robot System (MRS) Group*, Czech Technical University (CTU) in Prague, Prague, Czech Republic [RP5]
- Working with: Prof. Dr. Martin Saska.

Jun 2020 – Today **External Collaborator**, *Group for Research on Automatic Control Engineering (GRACE)*, University of Sannio (UniSannio), Benevento, Italy [RP6]
- Working with: Prof. Dr. Luigi Iannelli.

Jun 2020 – Jul 2022 **Postdoctoral Research Fellow**, *MRS Group*, Czech Technical University in Prague, Prague, Czech Republic [RP5], [M1]
- Supervisor: Prof. Dr. Martin Saska.

Dec 2019 – Jun 2020 **Research Fellow**, *Group for Research on Automatic Control Engineering (GRACE)*, University of Sannio, Benevento, Italy [RP6], [RP8]
- Supervisor: Prof. Dr. Luigi Iannelli.

May 2016 – Nov 2016 **Research Fellow**, *Group for Research on Automatic Control Engineering (GRACE)*, University of Sannio, Benevento, Italy
- Supervisor: Prof. Dr. Luigi Iannelli.

Professional Experience

Jun 2014 – Dec 2025 **Technical Writer**, *Writer of technical articles for magazines and web platforms*, i.e., "`Win Magazine`", "`Ridble`", "`Firmware`", "`EOS Book`", and "`Fare Elettronica`", Italy

Apr 2016 **Junior Software Engineer**, *Software Engine S.r.l.*, Solofra, Italy
- Web Developer (specialized in front-end development) responsible for the care (maintenance, debugging) and implementation of HTML5 and CSS3 templates (following the mobile-first responsive and cross browsing), javascript (jQuery, Angular-js) and back-end integration (Java), as well as, management and implementation of relational databases and debugging.
- Projects completed:
  - *Software developer for a management system*. Develop of a software for managing documents presented under the law 219/81 and ss.mm.ii. for the town of Mirabella Eclano, Italy.

Aug 2012 – Jan 2013 **Control System Integrator**, *Mosaico Monitoraggio Integrato S.r.l.*, Benevento, Italy
- Design and implementation of control systems for industrial automation based on PLC and Supervisory Control And Data Acquisition (SCADA).
- Projects completed:
  - *Storage and loading system of soda autoclaves in autoclaving*. Design and development of the system, and analysis of the control system requirements in accordance to the safety functions.
  - *Leaching process of the blades core for high performance turbines*. Design and development of the system, and analysis of the control system requirements in accordance to the safety functions.

Miscellaneous

Research Areas: Robotics; Uncrewed Aerial Vehicles (UAVs) control and navigation; software and simulation-in-the-loop; Temporal Logic (TL); robot manipulators; Model Predictive Control (MPC); communication-aware robotics; human-robot collaboration; formal methods in robotics and automation; autonomous systems; middleware for laboratory research infrastructures.

Skills and expertise: **Programming language**: C, C++, Java, Python. **Miscellaneous**: Jekyll, CMake, MATLAB/Simulink. **Robotics Middleware and 3D dynamic simulation environments**: Robot Operating System (ROS), Generator of Modules (GenoM), Gazebo, CoppeliaSim (formerly V-REP), MATLAB Virtual Reality Toolbox. **Markup languages**: Markdown, reStructuredText (RST), LaTeX. **Distributed version control systems**: Mercurial, Git. **Continuous integration systems**: Jenkins, GitHub Actions, TravisCI. **Optimization tools**: Gurobi, CasADi, CVX, IPOPT, qpOASES, YALMIP, MATMPC, Optimization Engine (OpEn), CPLEX. **Containerization platforms**: Docker.

License: **Aircraft Pilot's License**, Remote Piloting Media Vehicles Regulation, VL/Mc class, Aeroclub of Benevento Gen. Nicola Collarile, Benevento, Italy, April 2017

Open-source Dev.: Contributor, author, and maintainer of open-source projects:

- Among the authors and maintainer of the TeX repository "`europecv`". Comprehensive TEX Archive Network (CTAN) profile available at `https://ctan.org/home/g.silano`.
- Author of the CrazyS, BebopS, MAT-Fly packages.
- Contributor of the styles, RotorS, bebop_autonomy, NMPC strategy for Autonomous Vehicles, Vision-Based Path Following Algorithm, ros-travis-integration, kaobook, multi-robot inspection and maintenance software architecture, PACNav, JRA-3.1-JaNDER-API, and MATMPC packages, templates, and draw examples in LaTeX.
- GitHub and Gist profiles: `https://github.com/gsilano` and `https://gist.github.com/gsilano`, respectively.

Research Activities

Motion planning algorithms for the inspection of civilian infrastructure with multi-robot systems

Within the Aerial-Core European project [RP5][2], path planning algorithms aimed to compute feasible and constrained trajectories for a fleet of UAVs leveraging on Signal Temporal Logic (STL) specifications have been investigated. The planner allows to formulate rather complex missions avoiding obstacles and forbidden areas along the path when also considering vehicle constraints (e.g., maximum velocity and acceleration) [W6], [W10][2], [J8], and [J12]. Besides, a multi-layer software architecture to perform cooperative missions with a fleet of UAVs was designed [C18], [J9], and several miscellaneous contributions came out around the research activity in [RP5], such as [C12], [C13], [C15], [W5], [W9], [J5], [W4] and [J4]. A thorough analysis of existing robotics simulators was conducted [J7].

[2]The research activity was partially carried out also within the Comp4Drones European project [RP6].

Communication-aware robotics

The research activity focuses on "*communication-aware robotics*", which addresses the challenges of maintaining connectivity and robust information flow in robotic networks. With the increasing importance of robotic networks in various applications, such as emergency response and surveillance, wireless communication plays a crucial role in connecting UAVs with other nodes or remote operators. The main motivation behind this research is to develop methods and algorithms that enable UAVs to explicitly consider communication link qualities during control and motion planning. In this context, several frameworks were designed to generate feasible dynamic trajectories for multi-rotor UAVs relay while considering their motion dynamics and the motion-induced by changes of the antenna orientation was also proposed [C3], [C4], [C10], [C16]. Considerations where also made on the presence of jammers and eavesdroppers [P2], [J1], [W3], [J6], [C1], [C5], [C8].

Model Predictive Control strategies for motion control of aerial robots

Within the Aerial-Core European project [RP5], Model Predictive Control (MPC) strategies for vision-based target tracking and collision avoidance in multi-rotor aerial vehicles have been investigated. These proposed control strategies take into account realistic actuation limits, visual perception, and obstacle avoidance constraints [C14]. The research also addressed robust control of cooperative aerial structures under parametric uncertainty [C2], [W1].

Aerial robots formation control

Within the Aerial-Core European project [RP5], research activities focused on approaches for achieving decentralized collective navigation of UAV swarms. Simulated and real-world experiments conducted in a natural forest demonstrated the validity and effectiveness of the proposed collective navigation approach in challenging environments [J10], [W5]. Additionally, a comprehensive analysis of existing robotics simulators was carried out [J7].

Modeling and control of autonomous vehicles

It has been investigated a MPC strategy aimed at controlling a small-scalecar model for autonomous racing competitions. An optimization problem was set up considering both vehicle's actuation limits and the lateral and longitudinal forces acting on the car described by the Pacejka's Magic Formula and a drivetrain model. The approach allows to safely race on a track populated by static obstacles generating collision-free trajectories and tracking them while enhancing the lap timing performance. The feasibility and effectiveness of the control architecture are demonstrated through closed-loop simulations achieved in Gazebo, specifically using the F1/10 simulator [C11][2].

Electric Vehicles Integration into Smart Grids

Within the FLOW European project [RP3], the research focused on developing innovative strategies for integrating Electric Vehicles (EVs) into the energy

energy grid to improve flexibility and efficiency. The work aims to optimize EV deployment, promote sustainable energy management, and advance the seamless integration of EVs within smart grid infrastructures [J3].

Physical human-aerial robot interaction

Robots are designed to assist humans in physically demanding tasks and enhance their mental and physical well-being. Collaboration between humans and robots is crucial for tackling complex tasks, particularly in unstructured environments. However, human-aerial robot interaction remains largely unexplored due to the potential dangers that aerial vehicles pose to humans. To address this, a method has been developed that uses high-level specifications expressed in the form of a Signal Temporal Logic (STL) formula to automate task assignments and trajectories for a multi-rotor aerial vehicle performing object handovers in a power line setting. This method also considers human comfort factors such as ergonomics and preferences [J2], [C9], [W2], [W8][3], [A1]. Additionally, multi-UAV formation has been investigated to further assist human workers in such critical environments [J4], [W7]. A comprehensive analysis of existing robotics simulators was also conducted [J7].

Control algorithms for autonomously manage production and coordinate agricultural operations

Within the AFarCloud European project [RP8], control techniques aimed to autonomously managing the production and coordinating farming operations have been investigated, i.e., hierarchical planning algorithms (e.g., RRT, A*, etc.). Specifically, algorithms to coordinate the operations within the farming autonomous activity, thus enhancing productivity and cost reduction, gradually shifting towards a high-level autonomy [C17][2], [C20], and [J11][2].

Flight control of Uncrewed Aerial Vehicles

Various activities dealing with UAVs, in particular multi-rotor vehicles, [J13][2], [BC4], [C17][2], [C19], [C21] and [C22] have been carried out. Software-in-the-loop (SIL) simulation has been investigated as an effective tool for analyzing the behavior of the closed control loop [BC4] and for helping in the design of control strategies [C19].

- **CrazyS**: CrazyS is an extension of the ROS package RotorS, aimed to model, develop and integrate the Crazyflie 2.0 nano quad-rotor in the physics based simulation environment Gazebo. This simulation platform allows to understand quickly the behavior of the flight control system by comparing and evaluating different indoor and outdoor scenarios, with a details level quite close to reality. The proposed extension was developed using Ubuntu 16.04 and the Kinetic Kame version of ROS, but it is also fully compatible with Ubuntu 18.04 along with the Melodic Morenia distribution of ROS and Ubuntu 20.04 along with the Noetic Ninjemys release of ROS. The platform expands the RotorS capabilities by considering the Crazyflie 2.0 physical model, its flight control system and the Crazyflie's onboard Inertial Measurement Unit (IMU), as well. The code has been released as open-source under Apache license (`https://github.com/gsilano/CrazyS`)

[3]The research activity was partially carried out within the Research Visiting period spent at the University of Twente and funded by the European H2020 TERRINet project (grant agreement no. 730994), call no. 10.

with the aim to share the obtained results with other researchers who already use such tools and would like to use the platform [BC4], [C21].

- **BebopS**: The repository contains the ROS code developed for the "Aerial robotics control and perception challenge", the Industrial Challenge of the 26th Mediterranean Conference on Control and Automation (MED'18) [A4]. The code aims to simulate the dynamics of the Parrot Bebop 2 together with the flight controller (both high and low level) when external disturbances (e.g., wind gusts) acting on it. The control algorithms were designed in MATLAB/Simulink and validated in Gazebo by using the MathWorks Robotics System Toolbox (MRST). Therefore, the repository contains both ROS nodes and launch files needed to simulate the drone behavior when a tracking algorithm is run. Also, on GitHub the glue code written during the challenge to tackle the tracking problem with the real platform is available. Such code allows to send control commands and to receive data from the aircraft and the Motion Capture (MoCap) systems, i.e., Vicon. Finally, the repository contains the developed ROS code to connect the control algorithm to the Parrot Sphinx simulator including the Parrot Bebop onboard firmware in the loop. In this way, details like synchronization, timing issues, fixed point computation, overflow, divisions-by-zero, can be isolated and investigated in detail [C19]. The code has been released as open-source under Apache license at the link `https://github.com/gsilano/BebopS`.
- **MAT-Fly**: MAT-Fly is a numerical simulation platform for multi-rotors aircraft characterized by the ease of use and control development. The platform is based on MATLAB and the MathWorks Virtual Reality Toolbox (MVRT) that work together to simulate the behavior of a drone in a 3D environment while tracking a car that moves along a nontrivial path. The MVRT toolbox has been chosen due to the familiarity that students have with MATLAB and because it allows to move the attention to the classifier, the tracker, the reference generator and the trajectory tracking control thanks to its simple structure. The overall architecture is quite modular so that each block can be easily replaced with others by simplifying the development phase and by allowing to add even more functionalities. The code is released under Apache license and made available at the links `https://github.com/gsilano/MAT-Fly` and `https://www.mathworks.com/matlabcentral/fileexchange/72486-mat-fly`, thus making the numerical platform available for scientific and educational activities [J13], [C22].

Mohamed Bin Zayed International Robotics Challenge 2020

The Mohamed Bin Zayed International Robotics Challenge (MBZIRC) is a biennial international robotics competition that provides an ambitious and technologically demanding set of challenges and is open to all teams from all countries. Also, the MBZIRC competition aims to inspire future robotics through innovative solutions and technological excellence. Within the competition, specifically the "challenge 2" of the 2020 edition, autonomous aerial and ground robots have been carried out for navigation and manipulation tasks, in unstructured, outdoor and indoor environments. In particular, a

team of UAVs and a Uncrewed Ground Vehicle (UGV) collaborating to autonomously locate, pick, transport and assemble different types of brick shaped objects to build pre-defined structures, in an outdoor environment. The challenge is motivated by construction automation and autonomous robot based 3D printing of large structures. This research activity has been carried out during the visiting period spent at LAAS-CNRS in Toulouse [A3], [W11].

Services for research infrastructure integration, coupling, and automation

A driving force behind the realization of a sustainable energy supply is the integration of renewable energy resources. Due to their stochastic generation behavior, energy utilities are confronted with the challenge of managing more complex power grids. Additionally, technological advancements, controllable loads, integration with other energy sources, changing regulatory rules, and market liberalization require the adaptation of the system's operation. Proper operational concepts and intelligent automation provide the foundation to transform the existing power system into an intelligent entity known as a smart grid. Within the context of the H2020 ERIGrid 2.0 European project [RP2], tools and methodologies have been designed to facilitate the integration, coupling, and automation of renewable energy resources [BC1], [BC2], [BC3], [C6], [C7].

Research Projects

[RP1] **Participation within the research fund for the Italian Electrical System**, *Resilience and Security of the Energy System project*, financed under the contract agreement between RSE S.p.A. and the Ministry of Environment and Energy Security, Project Advisor for the Ricerca sul Sistema Energetico (RSE) S.p.A.: Eng. Giovanni Pirovano and Eng. Riccardo Lazzari, Project amount: 18.6 MEuros (only related to RSE activities), Role: leader from RSE's Department side (WP3 – LA3.01, LA3.04, and LA3.07 leader), January 2025 – Today

[RP2] **Participation within H2020-2019-1-RIA - INFRAIA**, *European Research Infrastructure supporting Smart Grid Systems Technology Development, Validation and Roll Out (ERIGRID2)*, research project agreement no. 870620, Project Advisor for the Ricerca sul Sistema Energetico (RSE) S.p.A.: Eng. Riccardo Lazzari, Project amount: 561 kEuros (only related to RSE activities), Role: work package leader (WP13 – JRA4), October 2022 – April 2025

[RP3] **Participation within HORIZON-CL5-2021-D5-01-RIA**, *Flexible energy systems Leveraging the Optimal integration of EVs deployment Wave (FLOW)*, research project agreement no. 101056730, Project Advisor for the Ricerca sul Sistema Energetico (RSE) S.p.A.: Eng. Filippo Colzi, Project amount: 458 kEuros (only related to RSE activities), Role: leader from RSE's Department side (T4.3 leader), July 2022 – June 2026

[RP4] **Participation within the research fund for the Italian Electrical System**, *Vehicle-to-Grid (V2G) project*, financed under the contract agreement between RSE S.p.A. and the Ministry of Economic Development - General Directorate for the Electricity Market, Renewable Energy and Energy Efficiency, Nuclear Energy, Project Advisor for the Ricerca sul Sistema Energetico (RSE) S.p.A.: Eng. Filippo Colzi and Eng. Riccardo Lazzari, Project amount: 8.4 MEuros (only related to RSE activities), Role: leader from RSE's Department side (WP2 – LA2.01, LA2.04, and LA2.07 leader), July 2022 – December 2024

[RP5] **Participation within H2020-2019-2-RIA-two stages - ICT**, *AERIAL COgnitive integrated multi-task Robotic system with Extended operation range and safety (AERIAL-CORE)*, research project agreement no. 871479, Project Advisor for the Czech Technical University in Prague: Prof. Dr. Martin Saska, Project amount: 461 kEuros (only related to CTU activities), Role: leader from CTU's side (T2.3 and T7.4 leader), June 2020 – July 2022, and participant, July 2022 – December 2023

[RP6] **Participation within H2020-2018-2-RIA-two-stages – ECSEL**, *Software Components For Drones (Comp4Drones)*, research project agreement no. 826610, Project Advisor for the University of Sannio: Prof. Dr. Luigi Iannelli, Project amount: 285 kEuros (only related to University of Sannio activities), Role: participant, October 2019 – May 2023

[RP7] **Participation within the Mohamed Bin Zayed International Robotic Challenge (MBZIRC) 2020**, Project Advisor: Prof. Dr. Antonio Franchi, Role: member of the Laboratoire d'Analyse et d'Architecture des Systemes (LAAS) team, March 2019 – February 2020

[RP8] **Participation within H2020-2017-2-RIA-two-stages – ECSEL**, *Aggregate Farming in the Cloud (AFarCloud)*, research project agreement no. 783221, Project Advisor for the University of Sannio: Prof. Dr. Luigi Glielmo, Role: leader from UniSannio side (T3.2 leader), September 2018 – November 2021

Research Visits

2022 November 17 — November 30, 2022, University of Twente, Enschede, The Netherlands, Robotics and Mechatronics (RaM) group, Prof. Dr. Antonio Franchi, funded by the European H2020 TERRINet project (grant agreement no. 730994), call no. 10.

Scientific Activity[4]

Editorial Board

2026 – 2028 Associate Editor of the IEEE Transactions on Automation Science and Engineering (T-ASE).

2025 – 2027 Associate Editor of the IEEE Control Systems Society (CSS) Conference Editorial Board.

2025 – 2026 Associate Editor of the IEEE International Conference on Robotics & Automation (ICRA)

[4]Link to the WebOfScience profile: `https://www.webofscience.com/wos/author/record/1668893`.

2025 Associate Editor of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

2024 – 2026 Early Career Editorial Member for Biomimetic Intelligence and Robotics (BIRob)

2025 Associate Editor of the IEEE International Conference on Systems, Man, and Cybernetics (SMC)

2022 – 2026 Associate Editor of the International Conference on Unmanned Aircraft Systems (ICUAS)

2024 – 2025 Associate Editor of the Mediterranean Conference on Control and Automation (MED)

2023 – 2024 Member of the Track Program Committee of the Robotics: Science and Systems (RSS). Role: reviewer.

2023 Member of the Track Program Committee of the 2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE'23). Member of the Technical Track Committee "TT 03 - Power Systems and the Smart Grid, Renewable Energy Systems and Smart Grid". Role: reviewer.

Expert Reviewer for Funding Agencies

2026 Remote Evaluator for European Innovation Council (EIC Transition – Horizon Europe), a flagship EU programme supporting breakthrough innovation.

2025 – 2026 Member of the Register of Evaluation Experts (EIV) for ANVUR (Italian National Agency for the Evaluation of Universities and Research Institutes). Tasked with the peer review of research outputs (e.g., articles, monographs, patents) within the Expert Evaluation Panel (GEV) for GEV 09 - Industrial and Information Engineering, in accordance with the Agency's criteria and guidelines.

2025 Registered expert in the REPRISE register (Register of Scientific Experts) established by the Italian Ministry of University and Research (MIUR), under the section Competitive Industrial Research and Social Development.

2025 Reviewer for the Agence Nationale Recherche (ANR), one of the leading governmental science funding organizations in France, supporting innovation and quality in research.

2025 Reviewer for the Dutch Research Council (NWO), one of the leading governmental science funding organizations in the Netherlands, supporting innovation and quality in research.

Scientific Societies

2023 – Today Member of the IEEE Robotics and Automation Society (RAS) Technical Committee on Smart Buildings

2023 – Today Member of the IEEE Robotics and Automation Society (RAS) Technical Committee on Algorithms for Planning and Control of Robot Motion

2023 – Today Member of the IEEE Robotics and Automation Society (RAS) Technical Committee on Aerial Robotics and Unmanned Aerial Vehicles

2023 – Today Member of the IEEE Robotics and Automation Society (RAS) Technical Committee on Space Robotics

2022 – Today Member of the IEEE Robotics and Automation Society (RAS) Technical Committee on Multi-Robot Systems

Edited Books Reviewer

2020 Koubaa, Anis (Ed.), "Robot Operating System (ROS) - The Complete Reference (Volume 6)", Springer International Publishing, 2021.

2019 Koubaa, Anis (Ed.), "Robot Operating System (ROS) - The Complete Reference (Volume 5)", Springer International Publishing, 2020.

2018 Koubaa, Anis (Ed.), "Robot Operating System (ROS) - The Complete Reference (Volume 4)", Springer International Publishing, 2019.

International Journals Reviewer

2026 IEEE Transactions on Neural Networks and Learning Systems

2026 Science Robotics

2026 Advanced Robotics Research

2026 MPDI Drones

2026 ICUAS Association Unmanned Aviation eMagazine

2025 IEEE Open Journal of the Communications Society

2025 – Today IEEE Systems, Man, and Cybernetics Magazine

2025 IEEE Wireless Communications

2025 – Today The International Journal of Robotics Research (IJRR)

2025 IEEE Communications Letters

2025 – Today IEEE Robotics & Automation Magazine (RAM)

2025 – Today IEEE Transactions on Automation Science and Engineering (T-ASE)

2025 IEEE Transactions on Intelligent Transportation Systems (T-ITS)

2025 IEEE Transactions on Internet of Things Journal (IoT-J)

2025 IEEE/ASME Transactions on Mechatronics (T-MEC)

2024 Biomimetic Intelligence and Robotics

2024 Computational and Structural Biotechnology Journal (CSBJ)

2024 – Today Engineering Applications of Artificial Intelligence (EAAI)

2024 IEEE Control Systems Letters (L-CSS)

2024 IEEE Open Journal of Control Systems (OJ-CSYS)

2024 IEEE Open Journal of the Industrial Electronics Society (OJIES)

2024 IEEE Open Journal of Vehicular Technology (OJVT)

2024 – Today Robotics and Autonomous Systems (RAS)

2024 – Today IEEE Transactions on Industrial Informatics (TII)

2024 IEEE Transactions on Intelligent Vehicles

2024 IEEE Transactions on Systems, Man and Cybernetics: Systems

2024 – Today IEEE Transactions on Robotics (T-RO)

2023 Archives of Electrical Engineering

2023 – Today Control Engineering Practice

2023 – Today Expert Systems With Applications

2023 ISA Transactions
2023 PLoS ONE
2023 Springer Scientific Reports
2023 Taylor Francis Cybernetics and Systems
2023 IEEE Transactions on Circuits and Systems for Video Technology
2022 – Today Automatica
2022 – Today Field Robotics
2022 IEEE Transactions on Automatic Control (TAC)
2022 IEEE Vehicular Technology Magazine
2022 Journal of Field Robotics
2021 IET Cyber-Systems and Robotics
2021 MDPI Energies
2021 Journal of Unmanned Vehicle Systems
2021 – Today IEEE Access
2021 – Today IEEE Robotics and Automation Letters (RA-L)
2019 – Today IEEE Transactions on Control Systems Technology (TCST)
2019 – Today Springer Journal of Intelligent & Robotic Systems (JINT)

International Conferences Reviewer

2025 IEEE Conference on Advanced Robotics and its Social Impact (ARSO)
2018 – Today IEEE American Control Conference (ACC)
2021 – Today IEEE International Conference on Systems, Man, and Cybernetics (SMC)
2021 – Today IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
2018 – Today IEEE Conference on Control Technology and Applications (CCTA)
2020 – Today International Conference on Unmanned Aircraft Systems (ICUAS)
2023 – Today Mediterranean Conference on Control and Automation (MED)
2021 – Today IEEE International Conference on Robotics and Automation (ICRA)
2020 – Today IEEE International Workshop on Metrology for Aerospace (MAS)
2019 – Today IEEE European Control Conference (ECC)
2019 – Today IEEE Conference on Decision and Control (CDC)
2025 IFAC Symposium on Mechatronic Systems
2024 IEEE TechDefense 2024
2023 International Conference on System Theory, Control and Computing (ICSTCC)
2023 – Today International Workshop on "Open Source Modelling and Simulation of Energy Systems" (OSMSES)
2023 International Federation of Automatic Control (IFAC)
2021 Workshop on Aerial Robotic Systems Physically Interacting with the Environment (AIRPHARO)

Conferences Services

Program Committees

2026 **Member of the Program Committee**, *Annual National Meeting of the SIDRA (Italian Society of Automatic Control), AUTOMATICA.IT 2026, University of Parma, September 2–4, 2026*, Website: `https://automatica2026.unipr.it/index.php/committees`

2025 **Tutorial Co-Chair**, *2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC), October 5-8, 2025, Vienna, Austria*, Website: `https://www.ieeesmc2025.org/committees.html`

Organized Tutorial Sessions

Jun 2026 **Organizer of the tutorial session "When Robotics Meet Wireless Communications: A Tale of Two Interconnected Worlds"**, *2026 International Conference on Unmanned Aircraft Systems (ICUAS'26), June 15 - 18, 2026, Corfu, Greece*, together with Prof. Dr. Daniel Bonilla Licea, Prof. Dr. Hajar El Hammouti, and Prof. Dr. Mounir Ghogho, Mohammed VI Polytechnic University, Ben Guerir, Morocco, and Prof. Dr. Martin Saska, Czech Technical University in Prague. Website: `https://uasconferences.com/2026_icuas/t2-half-day-workshop/`

Oct 2025 **Organizer of the tutorial session "Redefining UAV Wireless Networks: The Role of Fully-Actuated Multi-Rotor Systems in Robotics and Communication"**, *2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC'25), October 5 - 8, 2025, Vienna, Austria*, together with Prof. Dr. Daniel Bonilla Licea, Prof. Dr. Hajar El Hammouti, and Prof. Dr. Mounir Ghogho, Mohammed VI Polytechnic University, Ben Guerir, Morocco, and Prof. Dr. Martin Saska, Czech Technical University in Prague, Czech Republic. Website: `https://www.ieeesmc2025.org/tutorials.html`

Organized Workshops

May 2023 **Organizer of the workshop session "The Role of Robotics Simulators for Unmanned Aerial Vehicles"**, *2023 IEEE International Conference on Robotics and Automation (ICRA'23), May 29 - June 2, 2023, ExCeL London, UK*, together with Dr. Kimberly McGuire, Bitcraze AB, Malmo, Sweden, Prof. Dr. Chiara Gabellieri, Robotics and Mechatronics group, Faculty of Electrical Engineering, Mathematics & Computer Science, University of Twente in Enschede, The Netherlands, and Prof. Dr. Wolfgang Hönig, Intelligent Multi-Robot Coordination Lab of TU Berlin, Germany. Website: `https://imrclab.github.io/workshop-uav-sims-icra2023`

Organized Special Issues

Jul 2023 – Oct 2025 **Organizer of the special issue "Advanced Control and Path Planning of Unmanned Aerial Vehicles (UAVs)"**, *on MPDI Machines (ISSN 2075-1702), section "Mechatronic and Intelligent Machines"*, together with Prof. Dr. Yahui Liu, School of Vehicle and Mobility, Tsinghua University, Beijing, China. Website: `https://www.mdpi.com/journal/machines/special_issues/7ZS283VVX8`

Organized Invited Sessions

Jun 2024 **Organizer of the invited session "Aerial Robotics in Inspection and Maintenance Operations - From Design to Deployment"**, *2024 International Conference on Unmanned Aircraft Systems (ICUAS), June 4 - 7, 2024, Chania, Crete, Greece*, together with Prof. Dr. Chiara Gabellieri, Robotics and Mechatronics group, Faculty of Electrical Engineering, Mathematics & Computer Science, University of Twente in Enschede, The Netherlands, and Prof. Dr. Mario Selvaggio, Department of Electrical Engineering and Information Technology University of Naples Federico II, Naples, Italy. Website: `https://sites.google.com/view/icuas-2024-special-session/home`

Invited Talks

Nov 2024 **Exploring Laboratory Middleware for Distributed Simulations Across Research Infrastructures: A Case Study from the ERIGrid 2.0 Project**, *Invited Talk*, The 14th Mediterranean Conference on Power Generation Transmission, Distribution and Energy Conversion 2024 (MED POWER 2024). Panel no. 4 "Advanced simulation and laboratory methods in power systems", Athens, Greece, `https://medpower2024.org/panels/`

Sep 2024 **Are Robotic Simulators a Solved Problem? Criteria for Selection and Customization**, *Invited Seminar*, ROS Aerial Community, Online Meeting, `https://discourse.ros.org/t/september-2024-meetings-aerial-robotics-back-from-summer/39383`

Dec 2022 **Signal Temporal Logics as a Tool to Encode Complex Missions for Multi-Robot Aerial Systems**, *Invited Seminar*, University of Sannio in Benevento, Benevento, Italy

Apr 2016 **Advanced drone applications: opportunities and problems**, *Invited Talk*, Make & Share event, Tech Coffee association, Benevento, Italy

Chairman or co-chairman of International Conference Sessions

Oct 2025 2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC'25), Vienna, Austria. Role: chair.

Jun 2021 2021 International Conference on Unmanned Aircraft Systems (ICUAS'21), Athens, Greece. Role: chair.

Conference Contributions

Sep 2024 **2024 Open Source Modelling and Simulation of Energy Systems (OSMSES'24)**, *"Integrating Power-to-Heat Services in Geographically Distributed Multi-Energy Systems: A Case Study from the ERIGrid 2.0 Project"*, Vienna, Austria, Oral Presentation

May 2024 **2024 12th Workshop on Modelling and Simulation of Cyber-Physical Energy Systems (MSCPES'24)**, *"Laboratory Middleware for the Cyber-Physical Integration of Energy Research Infrastructures"*, Hong Kong, China (Virtual), Oral Presentation

Oct 2022 **2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC'22)**, *"A Nonlinear Model Predictive Control Strategy for Autonomous Racing of Scale Vehicles"*, Prague, Czech Republic, Oral Presentation

Sep 2021 **Automatica.it 2021**, *national meeting held by SIDRA, "Optimum Trajectory Planning for Multi-Rotor UAV Relays with Tilt and Antenna Orientation Variations"*, Catania, Italy, Oral Presentation

Jun 2021 **2021 International Conference on Unmanned Aircraft Systems (ICUAS'21)**, *"A Multi-Layer Software Architecture for Aerial Cognitive Multi-Robot Systems in Power Line Inspection Tasks"*, Athens, Greece, Oral Presentation

Jun 2021 **2021 IEEE International Conference on Robotics and Automation (ICRA'21)**, *"Power Line Inspection Tasks with Multi-Aerial Robot Systems via Signal Temporal Logic Specifications"*, Xi'an, China, Oral Presentation

Sep 2020 **Automatica.it 2020**, *national meeting held by SIDRA, "A framework for power line inspection tasks with multi-robot systems from signal temporal logic specifications"*, Cagliari, Italy, Oral Presentation

Jun 2018 **26th Mediterranean Conference on Control and Automation (MED'18)**, *"CrazyS: a software-in-the-loop platform for the Crazyflie 2.0 nano-quadcopter"*, Zadar, Croatia, Oral Presentation

Sep 2017 **Automatica.it 2017**, *national meeting held by SIDRA, "An educational simulation platform for Unmanned Aerial Vehicles aimed to detect and track moving objects"*, Milan, Italy, Poster Session

Teaching Experience

Teaching Activity

Jul 2024 **Teaching Activity for the PhD course "Fundamentals for Robot Programming with ROS" (1 ECTS credits)**, *PhD programme in Information Technologies for Engineering, Department of Engineering, University of Sannio*, Benevento, Italy, Academic Year 2023/2024

Teaching Assistance

Sep 2017 – Dec 2019 **Teaching Assistant for the graduate course "Discrete Systems" (9 ECTS credits)**, *first year of the Master degree (Laurea Magistrale) in Computer Engineering, Department of Engineering, University of Sannio*, Benevento, Italy, Academic Years 2017/2018 and 2018/2019, Supervisor: Prof. Dr. Luigi Iannelli

Mar 2016 – Jun 2019 **Teaching Assistant for the undergraduate course "Automatic Control" (6 ECTS credits)**, *second year of the Laurea degree (Laurea Triennale) in Computer and Electronics Engineering, Department of Engineering, University of Sannio*, Benevento, Italy, Academic Years 2016/2017 till 2018/2019, Supervisor: Prof. Dr. Luigi Iannelli

Sep 2018 – Dec 2018 **Teaching Assistant for the graduate course "Advanced Controls" (9 ECTS credits)**, *second year of the Master degree (Laurea Magistrale) in Electronics Engineering, Department of Engineering, University of Sannio*, Benevento, Italy, Academic Year 2018/2019, Supervisor: Prof. Dr. Luigi Glielmo

Subject Matter Expert ("Cultore della materia")

2017 – 2019 **Subject Matter Expert ("Cultore della materia")**, *University of Sannio*, Benevento, Italy, expert for the course "Sistemi Discreti" (art. 46 D.P.R. 28/12/2000 n. 445 of Italian law), under the instruction of Prof. Dr. Luigi Iannelli, Master's degree in Computer Science Engineering

2017 – 2018 **Subject Matter Expert ("Cultore della materia")**, *University of Sannio*, Benevento, Italy, expert for the course "Controlli Automatici" (art. 46 D.P.R. 28/12/2000 n. 445 of Italian law), under the instruction of Prof. Dr. Luigi Iannelli, Bachelor's degree in Electronics Engineering

Co-supervisor Scientific Activity

2020 **MIT Independent Activity Program of Ria Sonecha**, *"Software- and hardware-in-the-loop methodologies aimed to test and validate the open-source simulation platform CrazyS"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2018 **MIT Independent Activity Program of Benjamin Rodriguez**, *"Crazyflie 2.0 model and control for Gazebo 3D simulator"*, University of Sannio, Benevento, Italy, Advisors: Prof. Dr. Luigi Glielmo, Prof. Dr. Luigi Iannelli, and Prof. Dr. Davide Liuzza

Student Supervisor

2021 **Bachelor Thesis of Andriy Dmytruk**, *"A perception-aware NMPC for collision avoidance and control of a multi-rotor UAV with visual localization constraints"*, Ukrainian Catholic University, Ukraine, Advisor: Dr. Giuseppe Silano

2021 **Master Project Thesis of Giuseppe Caputo**, *"A framework to fulfill inspection missions for a multi-robot system via Linear Temporal Logic specifications"*, Czech Technical University in Prague, Czech Republic, Advisor: Dr. Giuseppe Silano

2021 **Master Project Thesis of Antonio Lombardo**, *"Signal Temporal Logic as a tool to encode inspection missions for a quad-rotor"*, Czech Technical University in Prague, Czech Republic, Advisor: Dr. Giuseppe Silano

Student Co-supervisor

2017 **Bachelor Thesis of Emanuele Aucone**, *"Hovering control for a nano quad-rotor"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2017 **Bachelor Thesis of Francesco Mariano D'Andrea**, *"Adaptive control of the vertical dynamics of a drone"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2017 **Bachelor Thesis of Olga Napolitano**, *"State estimation for a quad-rotor"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2017 **Bachelor Thesis of Susanna Maio**, *"Analysis of the vertical control of a quad-rotor in the V-REP simulation environment"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2016 **Bachelor Thesis of Andrea Mascia**, *"Study of the algorithm for the position estimation of a micro UAV"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2016 **Bachelor Thesis of Luca De Vincentis**, *"Experimental characterization of the flight control system of a micro UAV"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

Awards

[A1] **Winner of the Best Extend Abstract Award of the IEEE RAS TC on Aerial Robotics and Unmanned Aerial Vehicles (ARUAVs)**, *Workshop "Advancements in Aerial Physical Interaction" at IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2025*, Paper: "STL-Guided Trajectory Planning with Uncertainty-Aware Risk Assessment for Human–Aerial Robot Collaboration", USD 500 travel grant

[A2] **Among the participants of the UNISANNIO team, winner of the "MathWorks Minidrone Competition at IFAC2020"**, *International Robotics Competition*, organized by MathWorks, Berlin, Germany, Webpage: `https://bit.ly/2VLZZU0`, `http://tiny.cc/46p3pz`

[A3] **Among the participants of LAAS team [RP7], finalist of the "Mohamed Bin Zayed International Robotics Challenge (MBZIRC)"**, *International Robotics Competition*, organized by the Khalifa University, Abu Dhabi, United Arab Emirates, Webpage: `http://mbzirc2020team.laas.fr/`, `http://mbzirc.com/`, Video: `https://youtu.be/0uWvDba7Eug`

[A4] **Finalist of the "Aerial robotics control and perception challenge"**, *26th Mediterranean Conference on Control and Automation (MED'18) Industrial Challenge*, organized by the University of Zagreb, Croatia, ISSN: 1066–033X, DOI: `10.1109/MCS.2018.2888719`, URL: `http://dx.doi.org/10.1109/MCS.2018.2888719`

References

Prof. Dr. Luigi Iannelli, Department of Engineering, University of Sannio, Piazza Roma 21, 82100 Benevento, Italy, luigi.iannelli@unisannio.it

Prof. Dr. Martin Saska, Faculty of Electrical Engineering (FEL), Czech Technical University in Prague, Karlovo namesti 13, 12135, Prague 2, Czech Republic, martin.saska@cvut.cz

Prof. Dr. Antonio Franchi, Faculty of Electrical Engineering, Mathematics & Computer Science, University of Twente, Carré 3609, P.O. Box 217, 7500 AE Enschede, The Netherlands, a.franchi@utwente.nl

Prof. Dr. Davide Liuzza, Department of Engineering, University of Sannio, Piazza Roma, 21, 82100 Benevento, Italy, davide.liuzza@unisannio.it

Prof. Dr. Luigi Glielmo, Department of Electric Engineering and of Information Technologies, University of Naples, Via Claudio 21, 80125 Naples, Italy, luigi.glielmo@unina.it

Prof. Dr. Wolfgang Hönig, Institute of Computer Engineering and Microelectronics, EECS Faculty, TU Berlin, Marchstraße 23, 10587 Berlin, Germany, hoenig@tu-berlin.de

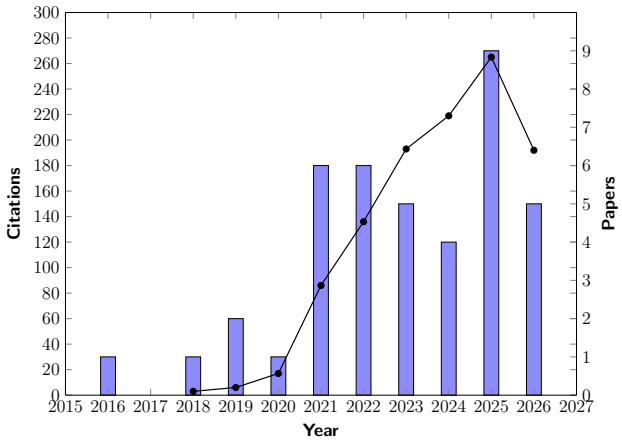
Prof. Dr. Mounir Ghogho, University Mohammed VI Polytechnic, Rabat Campus Rocade Rabat-Salé, 11103 Rabat, Morocco, mounir.ghogho@um6p.ma

Eng. Carlo Sandroni, Department of Power Generation Technologies and Materials, Ricerca sul Sistema Energetico, Via Raffaele Rubattino 54, 20134 Milan, Italy, carlo.sandroni@rse-web.it

Eng. Riccardo Lazzari, Department of Power Generation Technologies and Materials, Ricerca sul Sistema Energetico, Via Raffaele Rubattino 54, 20134 Milan, Italy, riccardo.lazzari@rse-web.it

Publications

The plot illustrates the progression of citations and the number of papers published over time, while the table below provides a summary of key citation metrics derived from Google Scholar. Additionally, I have had the privilege of **collaborating with over 86 unique co-authors**, with **17 recurring collaborators across ≥ 3 papers**, throughout my research career.



Citations Summary	
Total Citations	1133
h-index	15
i10-index	22

Publication Category	
Journal Articles	13
Conference Papers	22
Book Chapters	4
Workshop Papers	11
Preprints	22

Preprints

[P1] **G. Silano**. "*Readiness Barrier Functions: Forward-Invariant Control Authority for Overactuated Multirotor Allocation*", August 2026, ArXiv, DOI: 10.48550/arXiv.2608.16335, URL: http://arxiv.org/abs/2608.16335.

[P2] **G. Silano**, D. Bonilla Licea, D. Liuzza, A. Franchi, and M. Saska. "*Communications-Aware NMPC for Multi-Rotor Aerial Relay Networks Under Jamming Interference*", April 2026, ArXiv, DOI: 10.48550/arXiv.2603.28467, URL: https://arxiv.org/abs/2603.28467.

Referred Journals

[J1] A. K. A. H. Saliah, H. El Hammouti, D. Bonilla Licea, and **G. Silano**. "*Orientation-Aware Control and Trajectory Design for Aerial RIS-Assisted Wireless Communications*", IEEE Transactions on Vehicular Technology, vol. XX, no. XX, pp. XX-XX, Month Year, DOI: XX, URL: XX. IMPACT FACTOR: 7.5.

[J2] **G. Silano**, A. Afifi, M. Saska, and A. Franchi. "*STL-Based Motion Planning and Uncertainty-Aware Risk Analysis for Human-Robot Collaboration with a Multi-Rotor Aerial Vehicle*", Journal of Intelligent & Robotic Systems, vol. 112, no. 66, pp. 1-31, July 2026, DOI: 10.1007/s10846-026-02403-y, URL: http://doi.org/10.1007/s10846-026-02403-y. IMPACT FACTOR: 3.1.

[J3] D. Liuzza, E. Mostacciuolo, V. Mariani, L. Iannelli, F. Vasca, and **G. Silano**. "*Optimal Online Management of Vehicle-to-Grid Charging in Corporate Parking Facilities*", Sustainable Energy, Grids and Networks, vol. 46, art. no. 102205, pp. 1-22, June 2026, DOI: 10.1016/j.segan.2026.102205, URL: http://doi.org/10.1016/j.segan.2026.102205. IMPACT FACTOR: 5.6.

[J4] V. Kratky, **G. Silano**, M. Vrba, C. Papaioannidis, I. Mademlis, R. Penicka, I. Pitas, and M. Saska. "*Gesture-Controlled Aerial Robot Formation for Human-Swarm Interaction in Safety Monitoring Applications*", IEEE Robotics and Automation Letters, vol. 10, no. 8, pp. 8244-8251, August 2025, DOI: 10.1109/LRA.2025.3583607, URL: http://doi.org/10.1109/LRA.2025.3583607. IMPACT FACTOR: 5.3.

[J5] **G. Silano**, A. Caballero, D. Liuzza, L. Iannelli, S. Bogdan, and M. Saska. "*A Signal Temporal Logic Approach for Task-Based Coordination of Multi-Aerial Systems: a Wind Turbine Inspection Case Study*", Robotics and Autonomous Systems, vol. 186, art no. 104905, pp. 1-14, April 2025, DOI: 10.1016/j.robot.2024.104905, URL: http://doi.org/10.1016/j.robot.2024.104905. IMPACT FACTOR: 4.3

[J6] D. Bonilla Licea, **G. Silano**, H. El Hammouti, M. Ghogho, and M. Saska. "*Reshaping UAV-Enabled Communications with Omnidirectional Multi-Rotor Aerial Vehicles*", IEEE Communications Magazine, vol. 63, no. 5, pp. 94-100, 2025, DOI: 10.1109/MCOM.001.2400421, URL: http://doi.org/10.1109/MCOM.001.2400421. IMPACT FACTOR: 8.3.

[J7] C. A. Dimmig, **G. Silano**, K. McGuire, C. Gabellieri, W. Honig, J. Moore, and M. Kobilarov. "*Survey of Simulators for Aerial Robots: An Overview and In-Depth Systematic Comparisons*", IEEE Robotics and Automation Magazine, vol. 32, no. 2, pp. 153-166, June 2025, DOI: 10.1109/MRA.2024.3433171, URL: http://doi.org/10.1109/MRA.2024.3433171. IMPACT FACTOR: 7.1.

[J8] A. Caballero, and **G. Silano**. "*A Signal Temporal Logic Motion Planner for Bird Diverter Installation Tasks with Multi-Robot Aerial Systems*", IEEE Access, pp. 81361-81377, vol. 11, 2023, DOI: 10.1109/ACCESS.2023.3300240, URL: http://doi.org/10.1109/ACCESS.2023.3300240. IMPACT FACTOR: 3.9.

[J9] D. Hert, T. Baca, P. Petracek, V. Kratky, R. Penicka, V. Spurny, M. Petrlik, M. Vrba, D. Zaitlik, P. Stoudek, V. Walter, P. Stepan, J. Horyna, V. Pritzl, M. Sramek, A. Ahmad, **G. Silano**, D. Bonilla Licea, P. Stibinger, T. Nascimento, and M. Saska. "*MRS Drone: A Modular Platform for Real-World Deployment of Aerial Multi-Robot Systems*", Journal of Intelligent & Robotic Systems, pp. 1-34, vol. 108, no. 64, 2023, DOI: 10.1007/s10846-023-01879-2, URL: https://doi.org/10.1007/s10846-023-01879-2. IMPACT FACTOR: 3.3.

[J10] A. Ahmad, D. Bonilla Licea, **G. Silano**, T. Baca, and M. Saska. "*PACNav: A Collective Navigation Approach for UAV Swarms Deprived of Communication and External Localization*", Bioinspiration & Biomimetics, pp. 1-19, vol. 17, no. 6, 2022, DOI: 10.1088/1748-3190/ac98e6, URL: https://doi.org/10.1088/1748-3190/ac98e6. IMPACT FACTOR: 2.985.

[J11] A. Basiri, V. Mariani, **G. Silano**, M. Aatif, L. Iannelli, and L. Glielmo. "*A survey on the application of path-planning algorithms for multi-rotor UAVs in precision agriculture*", The Journal of Navigation, pp. 363-384, vol. 75, no. 2, 2022, DOI: 10.1017/S0373463321000825, URL: https://doi.org/10.1017/S0373463321000825. IMPACT FACTOR: 1.995.

[J12] **G. Silano**, T. Baca, R. Penicka, D. Liuzza, and M. Saska. "*Power Line Inspection Tasks with Multi-Aerial Robot Systems via Signal Temporal Logic Specifications*", IEEE Robotics and Automation Letters, vol. 6, no. 2, pp. 4169-4176, 2021, DOI: 10.1109/LRA.2021.3068114, URL: http://dx.doi.org/10.1109/LRA.2021.3068114. IMPACT FACTOR: 3.608. ACCEPTED ALSO TO ICRA'21.

[J13] **G. Silano**, and L. Iannelli. "*MAT-Fly: An Educational Platform for Simulating Unmanned Aerial Vehicles Aimed to Detect and Track Moving Objects*", IEEE Access, vol. 9, pp. 39333–39343, 2021, DOI: 10.1109/ACCESS.2021.3064758, URL: http://dx.doi.org/10.1109/ACCESS.2021.3064758. IMPACT FACTOR: 3.745.

Referred Book Chapters

[BC1] **G. Silano**, A. Kontou, Z. Feng, A. Acosta, O. Gehrke, T. A. Zerihun, S. Sanchez-Acevedo, G. Arnold, J.E. Rodriguez-Seco, R. Lazzari, C. Rodio, and L. Pellegrino. "Experiences with Smart System Integration and Validation". In European Guide to Smart Energy System Testing: The ERIGrid 2.0 Approach for Evaluating Complex Smart Energy System Configurations, Ed. by Thomas I. Strasser, Mihai Calin, Leonard Enrique Ramos Perez, SpringerBriefs in Energy, 2026, pp. 87-103, ISBN: 978-3-031-99451-7, DOI: 10.1007/978-3-031-99451-7_8, URL: https://doi.org/10.1007/978-3-031-99451-7_8.

[BC2] E. Widl, **G. Silano**, O. Gehrke, and T. A. Zerihun. "Sector Coupling and Multi-Domain Systems Validation". In European Guide to Smart Energy System Testing: The ERIGrid 2.0 Approach for Evaluating Complex Smart Energy System Configurations, Ed. by Thomas I. Strasser, Mihai Calin, Leonard Enrique Ramos Perez, SpringerBriefs in Energy, 2026, pp. 77-86, ISBN: 978-3-031-99451-7, DOI: 10.1007/978-3-031-99451-7_7, URL: https://doi.org/10.1007/978-3-031-99451-7_7.

[BC3] A. Acosta, **G. Silano**, G. Paludetto, O. Gehrke, M.C. Pham, Q.T. Tran, V. Rajkumar, S. Vogel, and A. Monti. "Laboratory Infrastructure Integration and Automation". In European Guide to Smart Energy System Testing: The ERIGrid 2.0 Approach for Evaluating Complex Smart Energy System Configurations, Ed. by Thomas I. Strasser, Mihai Calin, Leonard Enrique Ramos Perez, SpringerBriefs in Energy, 2026, pp. 65-76, ISBN: 978-3-031-99451-7, DOI: 10.1007/978-3-031-99451-7_6, URL: https://doi.org/10.1007/978-3-031-99451-7_6.

[BC4] **G. Silano**, and L. Iannelli. "*CrazyS: a software-in-the-loop simulation platform for the Crazyflie 2.0 nano-quadcopter*". In Robot Operating System (ROS): The Complete Reference (Volume 4), Ed. by Koubaa, Anis, Springer International Publishing, 2020, pp. 81–115, ISBN: 978-3-030-20190-6, DOI: 10.1007/978-3-030-20190-6_4, URL: https://doi.org/10.1007/978-3-030-20190-6_4.

Peer-Reviewed Conference Papers

[C1] M. Bajja, A. K. A. H. Saliah, H. El Hammouti, D. Bonilla Licea, and **G. Silano**. "*3D Euler-Angle Orientation Control for Two-Ray Fading Mitigation in Maritime Air-to-Sea Communications*", 2026 IEEE Global Communications Conference (IEEE GLOBECOM). Macau S.A.R., China, December 2026, pp. XX-XX, ISBN: XX, DOI: XX, URL: XX.

[C2] **G. Silano**, Q. Sablè, M. Tognon, L. Iannelli, and A. Franchi. "*Sensitivity-Based Tube NMPC for Cooperative Aerial Structures Under Parametric Uncertainty*", 2026 International Conference on Unmanned Aircraft Systems (ICUAS). Corfu, Greece, June 2026, pp. 73-79, ISBN: 979-8-3315-9316-2, DOI: 10.1109/ICUAS69441.2026.11598699, URL: https://doi.org/10.1109/ICUAS69441.2026.11598699.

[C3] **G. Silano**, D. Bonilla Licea, H. El Hammouti, and M. Saska. "*Free-Space Optical Communication-Driven NMPC Framework for Multi-Rotor Aerial Vehicles in Structured Inspection Scenarios*", 2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC). Vienna, Austria, October 2025, pp. 6641-6646, ISBN: 979-8-3315-3358-8, DOI: 10.1109/SMC58881.2025.11343117, URL: https://doi.org/10.1109/SMC58881.2025.11343117.

[C4] D. Bonilla Licea, **G. Silano**, and M. Saska, "*Enhanced Security and Coordination Framework for UAV Swarms Using Heterogeneous Communication Networks*", In: Andreoni, M., Thakkar, S. (eds) Proceedings of 1st GENZERO Workshop. GENZERO 2024, Hilton Yas, Abu Dhabi. Springer, Singapore, pp. 117–123, 2026, ISBN: 978-981-95-1050-4, DOI: 10.1007/978-981-95-1050-4_14, URL: https://doi.org/10.1007/978-981-95-1050-4_14.

[C5] D. Bonilla Licea, H. El Hammouti, **G. Silano**, and M. Saska. "*Harnessing the Potential of Omnidirectional Multi-Rotor Aerial Vehicles in Cooperative Jamming Against Eavesdropping*", 2024 IEEE Global Communications Conference (IEEE GLOBECOM). Cape Town, South Africa, December 2024, pp. 2052-2058, ISBN: 979-8-3503-5125-5, DOI: 10.1109/GLOBECOM52923.2024.10901802, URL: https://doi.org/10.1109/GLOBECOM52923.2024.10901802.

[C6] **G. Silano**, E. Rikos, V. Rajkumar, O. Gehrke, T. A. Zerihun, C. Rodio, and R. Lazzari. "*Integrating Power-to-Heat Services in Geographically Distributed Multi-Energy Systems: A Case Study from the ERIGrid 2.0 Project*", 2024 Open Source Modelling and Simulation of Energy Systems (OSMSES 2024). Vienna, Austria, September 2024, pp. 1-5, ISBN: 979-8-3503-8468-0, DOI: 10.1109/OSMSES62085.2024.10668976, URL: https://doi.org/10.1109/OSMSES62085.2024.10668976.

[C7] V. Rajkumar, **G. Silano**, O. Gehrke, S. Vogel, E. Widl, G. Paludetto, E. Rikos, T. A. Zerihun, A. Stefanov, P. Palensky, and T. I. Strasser. "*Laboratory Middleware for the Cyber-Physical Integration of Energy Research Infrastructures*", 2024 12th Workshop on Modelling and Simulation of Cyber-Physical Energy Systems (MSCPES). Hong Kong, China, May 2024, pp. 1-5, ISBN: 979-8-3503-6284-8, DOI: 10.1109/MSCPES62135.2024.10542755, URL: https://doi.org/10.1109/MSCPES62135.2024.10542755.

[C8] D. Bonilla Licea, **G. Silano**, M. Ghogho, and M. Saska. "*Omnidirectional Multi-Rotor Aerial Vehicle Poser Optimization: A Novel Approach to Physical Layer Security*", 2024 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP). Seoul, Korea, April 2024, pp. 9021-9025, ISBN: 979-8-3503-4485-1, DOI: 10.1109/ICASSP48485.2024.10447876, URL: https://doi.org/10.1109/ICASSP48485.2024.10447876.

[C9] **G. Silano**, A. Afifi, M. Saska, and A. Franchi. "*A Signal Temporal Logic Planner for Ergonomic Human–Robot Collaboration*", 2023 International Conference on Unmanned Aircraft Systems (ICUAS). Lazarski University, Warsaw, Poland, June 2023, pp. 328-335, ISBN: 979-8-3503-1037-5, DOI: 10.1109/ICUAS57906.2023.10156559, URL: https://doi.org/10.1109/ICUAS57906.2023.10156559.

[C10] D. Bonilla Licea, **G. Silano**, M. Ghogho, and M. Saska. "*Communications-Aware Robotics: Challenges and Opportunities*", 2023 International Conference on Unmanned Aircraft Systems (ICUAS). Lazarski University, Warsaw, Poland, June 2023, pp. 366-371, ISBN: 979-8-3503-1037-5, DOI: 10.1109/ICUAS57906.2023.10155882, URL: https://doi.org/10.1109/ICUAS57906.2023.10155882.

[C11] V. Cataffo, **G. Silano**, L. Iannelli, V. Puig, and L. Glielmo. "*A Nonlinear Model Predictive Control Strategy for Autonomous Racing of Scale Vehicles*", 2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC). Prague, Czech Republic, October 2022, pp. 100-105, ISBN: 978-1-6654-5258-8, DOI: 10.1109/SMC53654.2022.9945279, URL: https://doi.org/10.1109/SMC53654.2022.9945279.

[C12] D. Hert, T. Baca, P. Petracek, V. Kratky, V. Spurny, M. Petrlik, M. Vrba, D. Zaitlik, P. Stoudek, V. Walter, P. Stepan, J. Horyna, V. Pritzl, **G. Silano**, D. Bonilla Licea, P. Stibinger, R. Penicka, T. Nascimento, and M. Saska. "*MRS Modular UAV Hardware Platforms for Supporting Research in Real-World Outdoor and Indoor Environments*", 2022 International Conference on Unmanned Aircraft Systems (ICUAS). Drubovnik, Croatia. June 2022, pp. 1264-1273, ISBN: 978-1-6654-0593-5, DOI: 10.1109/ICUAS54217.2022.9836083, URL: https://doi.org/10.1109/ICUAS54217.2022.9836083.

[C13] A. Calvo, **G. Silano**, and J. Capitan. "*Mission Planning and Execution in Heterogeneous Teams of Aerial Robots supporting Power Line Inspection Operations*", 2022 International Conference on Unmanned Aircraft Systems (ICUAS). Drubovnik, Croatia. June 2022, pp. 1644-1649, ISBN: 978-1-6654-0593-5, DOI: 10.1109/ICUAS54217.2022.9836234, URL: https://doi.org/10.1109/ICUAS54217.2022.9836234.

[C14] A. Dmytruk, **G. Silano**, D. Bicego, D. Bonilla Licea, and M. Saska. "*A Perception-Aware NMPC for Vision-Based Target Tracking and Collision Avoidance with a Multi-Rotor UAV*", 2022 International Conference on Unmanned Aircraft Systems (ICUAS). Drubovnik, Croatia. June 2022, pp. 1668-1673, ISBN: 978-1-6654-0593-5, DOI: 10.1109/ICUAS54217.2022.9836071, URL: https://doi.org/10.1109/ICUAS54217.2022.9836071.

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[W8] **G. Silano**, A. Afifi, M. Saska, and A. Franchi, "*Ergonomic Collaboration between Humans and Robots: An Energy-Aware Signal Temporal Logic Perspective*". ICRA 2023 Energy Efficient Aerial Robotic Systems Workshop, ExCeL London, UK, June 2023, pp. 1-4, DOI: 10.48550/arXiv.2306.02454. URL: https://arxiv.org/abs/2306.02454.

[W9] **G. Silano**, V. Kratky, J. Bednar, M. Vrba, F. Nekovar, D. Bonilla Licea, T. Baca, P. Stoudek, D. Hert, M. Petrlik, D. Smrcka, T. Nascimento, and M. Saska, "*Coordination of a Heterogeneous Team of Aerial Robots to Support Power Line Inspection Operations*". 2022 International Conference on Unmanned Aircraft Systems (ICUAS). Drubovnik, Croatia. June 2022, contribution accepted for discussion at the tutorial session: "*AERIAL-CORE – Boosting the adoption of aerial robotics in real-world applications*", DOI: 10.1109/MCS.2022.3209088, URL: http://dx.doi.org/10.1109/MCS.2022.3209088.

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Posters

[P1] **G. Silano**, P. Oppido, and L. Iannelli, "*Software-in-the-loop simulation for improving flight control system design: a quadrotor case study*". In SIDRA National Meeting, Ancona, Italy, September 2019. DOI: 10.13140/RG.2.2.31583.61603, URL: http://dx.doi.org/10.13140/RG.2.2.31583.61603.

[P2] **G. Silano**, and L. Iannelli, "*An educational simulation platform for Unmanned Aerial Vehicles aimed to detect and track moving objects*". In SIDRA National Meeting, Milan, Italy, September 2017. DOI: 10.13140/RG.2.2.14878.43849, URL: http://dx.doi.org/10.13140/RG.2.2.14878.43849.

Miscellaneous

[M1] T. Prihodova, **G. Silano**, A. Ahmad, V. Kratky, T. Baca, P. Petracek, V. Saskova, J. Bednar, and M. Saska. "*2022 IEEE Robotics and Automation Society Summer School on Multi-Robot Systems in Prague [Education]*", IEEE Robotics and Automation Magazine, vol. 30, no. 1, pp. 104–106, 2023, DOI: 10.1109/MRA.2023.3238213, URL: https://doi.org/10.1109/MRA.2023.3238213. IMPACT FACTOR: 5.229.

Theses

Dec 2016 – Jul 2020 **Ph.D. Thesis**, *Software-in-the-loop methodologies for the analysis and control design of small UAV systems*, University of Sannio, Benevento, Italy

- Show the role and the effectiveness of robotic simulators and Software-in-the-loop methodologies in the flight control system design for multi-rotor vehicles.
- Advisors: Prof. Dr. Luigi Iannelli.

Sep 2015 – Mar 2016 **Master Thesis**, *Development of a simulator aimed to detect and track moving objects for UAVs*, University of Sannio, Benevento, Italy

- Design and develop of a 3D simulator aimed to detect and track arbitrary moving objects for UAVs.
- Advisors: Prof. Dr. Luigi Iannelli.

Feb 2012 – Jun 2012 **Bachelor Thesis**, *An Object Oriented approach aimed to the creation of control software for industrial processes*, Mosaico Monitoraggio Integrato S.r.l, Benevento, Italy

- Design a software production methodology using an Object Oriented (OO) approach for Programmable Logic Controller (PLC).
- Advisors: Prof. Dr. Luigi Iannelli and Eng. Paolo Rubino.